# Adham Ehab Erfan Mahmoud

Wireless Communications & RF Engineer | Embedded Systems & Automation

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## Professional Summary

Electronics & Communication Engineering graduate (GPA 3.5/4.0, *Excellent*) with hands-on experience in wireless system design, RF/embedded hardware, and AI-driven signal processing. Internship at **Siemens Egypt** covering AVR/ARM microcontrollers, serial protocols (UART, SPI, $I^2C$), and ESP32-based IoT data acquisition. Graduation project (A+) delivered a deep-learning beamforming framework for multi-cell 5G MIMO networks using ray-traced channel simulations. Seeking to apply physical-layer expertise and embedded programming skills to RF engineering, wireless network deployment, and industrial automation roles.

## Core Competencies

- MIMO & Antenna Systems
- 5G NR Physical Layer (SSB/CSI-RS)
- RF Link Budget Analysis
- OFDM Transceiver Design
- Channel Estimation & Equalisation
- Beamforming & Codebook Design
- Digital Signal Processing
- Spectrum & Propagation Analysis
- ARM / AVR Microcontrollers
- UART / SPI / $I^2C$ Protocols
- MATLAB / Simulink
- Python (Tensorflow, NumPy, Sionna)
- C / C++ Embedded Programming
- Git, LaTeX

## Industry Experience

**Embedded Systems Intern** — *Summer 2024*
*Siemens Egypt* — *Egypt*

- Worked with AVR and ARM Cortex-M microcontrollers: startup code, memory mapping, and low-level peripheral driver development.
- Implemented and debugged serial protocols (UART, SPI, $I^2C$) for sensor interfacing and inter-device communication.
- Programmed ESP32 for real-time data acquisition and wireless cloud streaming for remote industrial monitoring.
- Practiced schematic reading, hardware–software integration, and systematic debugging with oscilloscope and logic analyser.

**AI Model Developer** — *2024 – 2025*
*Outlier / Aligner* — *Remote*

- Developed and evaluated AI-generated outputs in mathematics and algorithm design; provided structured feedback to guide training pipelines.

**R&D Volunteer** — *2024 – Present*
*University Centre for Career Development (UCCD)* — *Alexandria, Egypt*

- Contributed to technical research documentation and engineering analysis reports.

## Education

**B.Sc. Electronics & Communication Engineering** — *Expected June 2026*
*Alexandria University — Faculty of Engineering* — *Alexandria, Egypt*

- GPA: **3.56 / 4.0** (*Excellent*) Graduation project: Neural Codebook Design for MIMO Beam Management
- Relevant coursework: Digital Communications, Wireless Networks, Antenna Theory, DSP, Control Systems, Embedded Systems, Machine Learning

## Key Technical Projects

**5G MIMO Neural Beamforming Framework** — *Wireless Communications / Deep Learning*

- Designed and trained a deep-learning model (PyTorch + Sionna) to generate SSB and CSI-RS codebooks for multi-cell 5G, maximising spectral efficiency under strict feedback overhead constraints.
- Simulated ray-traced multi-cell RF environments with overlapping BS coverage; modelled realistic multi-path propagation and inter-cell interference.
- Built a unified end-to-end pipeline: initial access, beam refinement, CSI feedback compression, and predictive beam tracking.
- Evaluated power allocation and interference trade-offs; *manuscript in preparation* for IEEE submission (2026).

**OFDM & 2×2 MIMO Physical-Layer Simulation** — *RF / Digital Communications*

- Implemented a full OFDM transceiver in MATLAB: pilot-based channel estimation, MMSE equalisation, and cyclic-prefix management.
- Extended to 2×2 MIMO with spatial multiplexing; measured BER across AWGN and Rayleigh fading channels over a wide SNR range.

**DQN-Based Frequency Jammer** — *Physical Layer Security / Spectrum Engineering*

- Trained a Deep Q-Network agent to adaptively target FHSS frequency bands with no prior knowledge of the hopping pattern; implemented experience replay, target networks, and $\varepsilon$-greedy exploration.

**Reconfigurable Intelligent Surfaces (RIS) for 5G/6G** — *RF / Antenna Systems*

- Analysed RIS phase-shift design strategies: compared alternating optimisation (AO) vs. deep-learning passive beamforming for SNR enhancement and coverage extension.

## Technical Skills

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|---|---|
| **RF & Wireless** | MIMO, OFDM, 5G NR (SSB/CSI-RS), Beamforming, Link Budget Analysis, Antenna Design, Propagation Modelling, Spectrum Analysis |
| **MATLAB** | Digital Comms Toolbox, DSP Toolbox, RF Toolbox, Simulink, Beamforming & Channel Simulation |
| **Python** | NumPy, PyTorch, TensorFlow, Sionna, MATLAB |
| **Embedded / HW** | C, C++, ARM Cortex-M, AVR, ESP32, UART/SPI/I$^2$C, PCB Design, IoT, RTOS basics, ROS |
| **AI / ML** | Deep Learning, CNNs, Autoencoders, DQN, Channel Estimation, Adaptive Algorithms |
| **Tools** | Git, LaTeX, MATLAB Simulink, Sionna, KiCad |
| **Languages** | Arabic (Native) \| English — Professional (IELTS Academic 6.0) |

## Certifications

**IELTS Academic** — Band 6.0 (British Council)

**Embedded Systems Essentials** — Siemens

**Deep Learning** — NVIDIA

**Machine Learning** — Generative Technology Center